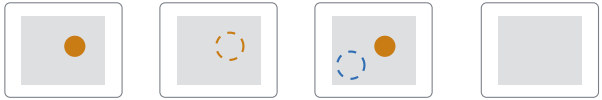
1 Counterfactual views, built without annotation



Expected answers: present / **absent** / present / absent. ESG = P(absent | erased) − P(absent | control): zero for any model that ignores the image or reacts to both edits alike.

2 A 4B agent trained on them

Qwen3.5-4B + LoRA
zoom / window tools
answer · refer · clarify
label-derived trajectories,
pre-registered selection

3 Tested once against 9 frontier and 6 open VLMs

Claims an erased lesion: **9.6%** vs 41–62% (frontier)

ESG: **0.76** vs 0.27–0.47; real-lesion accuracy not different

Holds under 6 prompts/settings, reworded questions, 2 unseen inpainting tools, a 2nd base model

Pre-registered checks: control matching not needed; removal-based DPO ties on ESG with a different error profile

Fresh external test: every pneumothorax missed (coverage gap)